
Healthcare Lead Generation MMM

PyMC-Marketing reference implementation for ROI, marginal returns, saturation and budget allocation

Project	Healthcare Lead Generation MMM with PyMC-Marketing
Data	24 months weekly, simulated but realistic
Channels	Google Search, Meta, YouTube, Display
KPI	Qualified calls and appointment bookings
Deliverables	Dataset, code, dashboard, methodology, report, proposal

Disclosure: This is a portfolio reference implementation using synthetic data. It is designed to demonstrate a production-minded MMM workflow without exposing client or patient data.

1. Executive Summary

This project demonstrates a first healthcare MMM build for a measurement startup with a paying client lined up. It is structured as a high-end proof of capability: realistic weekly data, media transformations, conservative inference, business-facing outputs, Streamlit dashboard and a full handoff package.

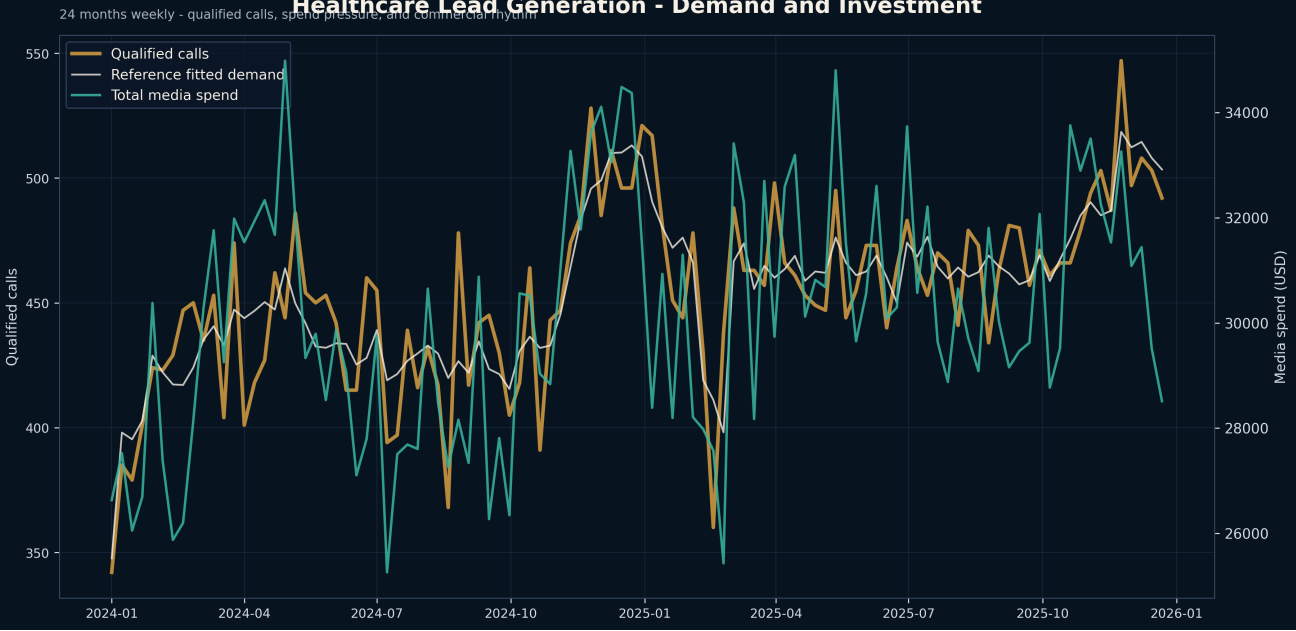
The primary KPI is qualified calls. This is more credible for healthcare lead generation than raw calls because it better represents commercial intent and filters out low-quality interactions. Appointment bookings are included as a secondary downstream conversion signal.

The model is designed for decision support under uncertainty. It estimates channel contribution, ROAS, marginal ROAS, saturation and a conservative budget reallocation scenario. It avoids the common error of presenting MMM as exact deterministic attribution.

Executive Performance Scorecard

Channel	Spend	Incremental Calls	Cost / Incr. Call	ROAS	mROAS
Google Search	\$1,357k	6,506	\$209	1.08x	2.15x
Meta	\$907k	4,266	\$213	1.06x	1.57x
YouTube	\$494k	3,914	\$126	1.78x	1.49x
Display	\$377k	5,800	\$65	3.46x	1.11x

Healthcare Lead Generation - Demand and Investment

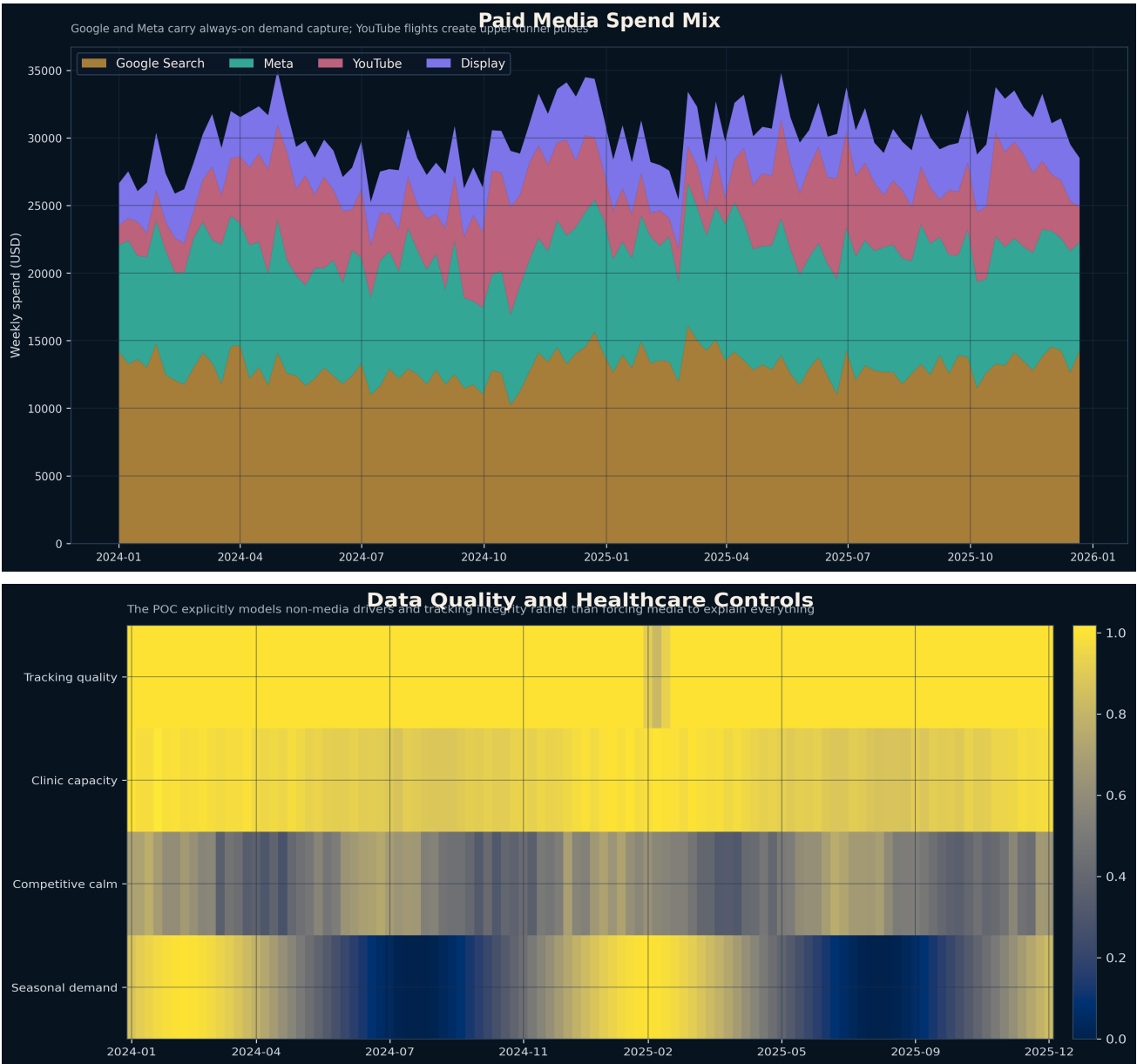


2. Dataset Design

The dataset contains 104 weekly observations, representing 24 months of healthcare lead-generation activity. It includes paid media investment, qualified calls, appointment bookings and healthcare-relevant controls.

The data was intentionally built with realistic MMM challenges: correlated Google and Meta spend, upper-funnel YouTube flying, always-on display support, seasonal demand, operational capacity variation, competitor pressure and a short call-tracking disruption.

This design makes the POC stronger than a clean toy dataset. It forces the model and dashboard to handle the same problems that appear in real healthcare measurement work.

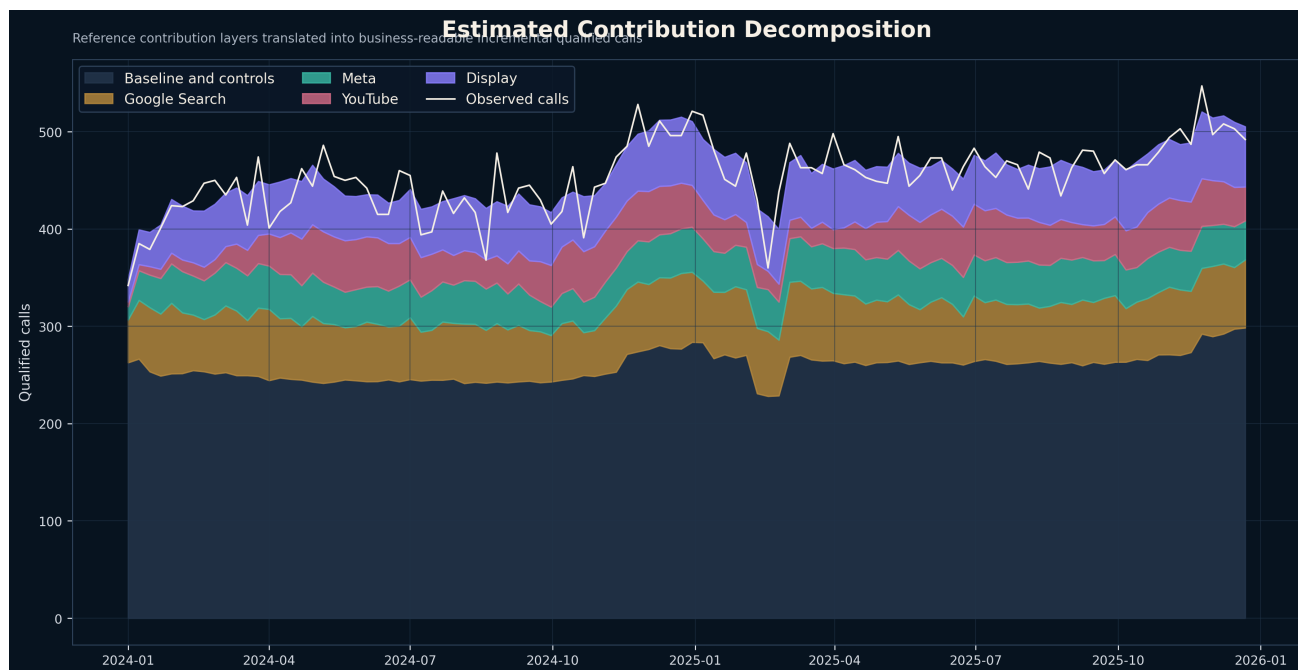


3. Model Logic

The model follows the standard MMM structure: baseline demand plus controls plus transformed media contribution. Each media channel passes through adstock and saturation before contribution is estimated.

Adstock captures delayed media impact. This is important because YouTube and display can influence demand after exposure, while search is usually more immediate. Saturation captures diminishing returns, which prevents the model from assuming that every additional dollar has the same impact.

The PyMC-Marketing implementation template is included in the repository. The generated outputs use a stabilized reference fit to produce portfolio-ready contribution, ROI and budget examples without requiring client infrastructure.

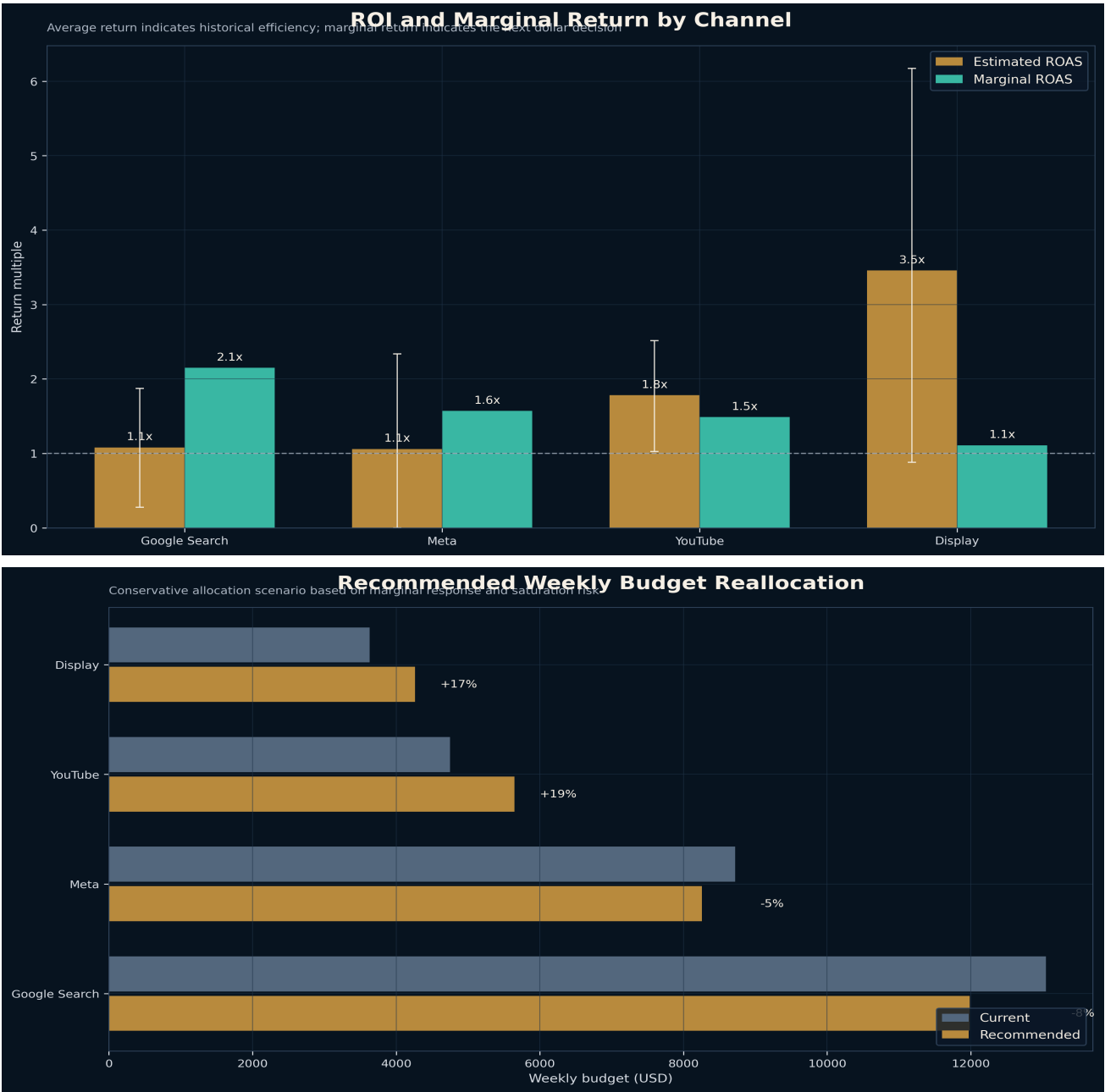


4. ROI, mROAS and Budget Interpretation

ROAS describes historical efficiency. Marginal ROAS describes the expected return of the next increment of spend. For budget decisions, marginal ROAS is usually more important than average ROAS.

The budget recommendation is intentionally conservative. It blends the current allocation with marginal response signals, rather than reallocating aggressively to the highest point estimate. This is more appropriate for a first healthcare model with limited historical data.

The dashboard and reports therefore separate measurement from decisioning: the model estimates contribution and uncertainty; the business layer translates the estimates into cautious action.

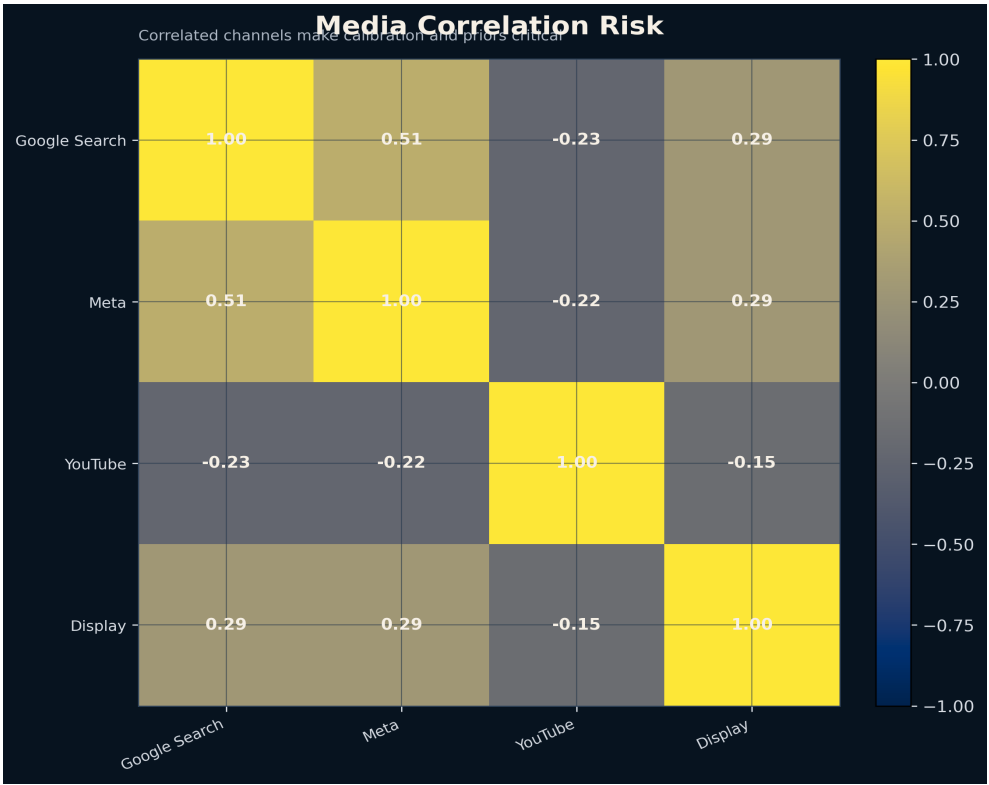
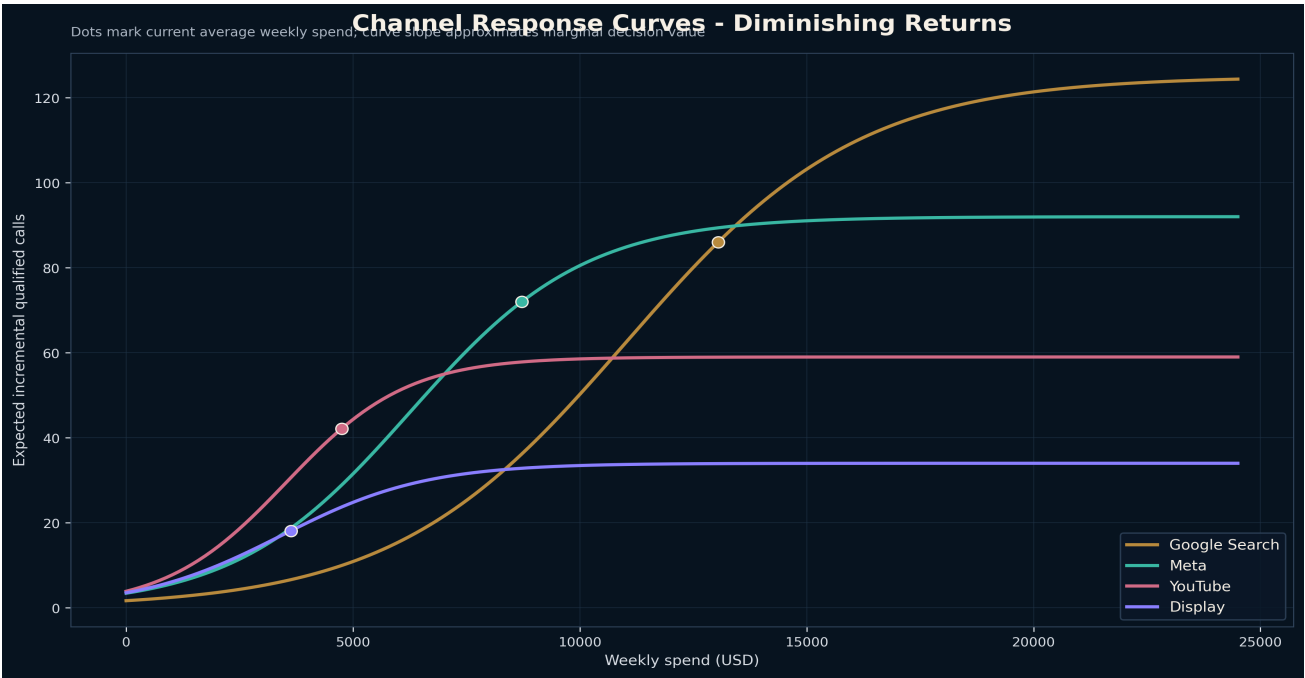


5. Saturation and Channel Scaling

Saturation curves show the expected relationship between weekly spend and incremental qualified calls. The curve slope is the practical signal: steep slope means attractive scaling potential; flat slope means incremental budget is likely to be inefficient.

The dots indicate current average weekly spend. This makes the chart immediately useful for business stakeholders because it connects model logic to current spending decisions.

The model should be recalibrated when the client introduces new offers, new markets, major budget changes or new conversion definitions.



6. Channel Summary

The table below summarizes the reference model output. In a real client engagement, this table should be presented with credible intervals, calibration notes and explicit caveats around correlated channels.

Channel	Spend	Inc. Calls	ROAS	mROAS	Cost / Inc. Call
Google Search	\$1,357k	6,506	1.08x	2.15x	\$209
Meta	\$907k	4,266	1.06x	1.57x	\$213
YouTube	\$494k	3,914	1.78x	1.49x	\$126
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7. Calibration Plan

The first client question should be whether there is any calibration signal: lift tests, geo tests, holdout periods, campaign experiments or prior business constraints on expected ROI by channel. This matters because 18-24 months of weekly data can be insufficient to separate highly correlated channels without priors or calibration.

If no experiments exist, the first version should rely on domain-informed priors, posterior predictive validation, residual checks, spend-response curve review and business plausibility checks. The report should clearly distinguish calibrated findings from uncalibrated model estimates.

8. Limitations

This project uses simulated data and should be presented as a reference implementation, not as real client evidence. In healthcare, a production model should also validate HIPAA-compliant data handling, conversion definitions, call quality logic, appointment capacity and tracking continuity.

The correct business language is estimated contribution, expected marginal return and decision support under uncertainty. The model should not claim exact user-level attribution or definitive causality without experimental support.

9. Final Positioning

This project is designed to support a premium Upwork positioning: Bayesian MMM Engineer for Healthcare Marketing Measurement. It demonstrates technical competence, business translation and methodological caution.

The strongest commercial message is simple: I can build the first model quickly, but I will not overclaim its precision. I will make assumptions explicit, calibrate where possible, and deliver outputs that executives can actually use.

Included deliverables: dataset, source code, notebooks, Streamlit app, figures, outputs, methodology documents, executive PDF report and Upwork proposal.